Modifiable risk factors drive a large share of the global cancer burden

Using global cancer incidence data from 185 countries, this study shows that four in ten cancers worldwide are attributable to modifiable risk factors. The findings highlight substantial opportunities for prevention through targeted, population-level interventions adapted to regional and sex-specific risk profiles.

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The problem

A substantial proportion of cancer burden is preventable through reductions in modifiable risk factors[1–3]. Advances in treatment have improved survival for some cancers, but population-level declines in cancer mortality have largely been driven by prevention and early detection[4]. Persistent global and regional inequalities in cancer incidence reflect marked differences in exposure to behavioral, environmental, occupational and infectious risk factors, many of which are amenable to policy intervention[1,2]. However, comprehensive and comparable estimates of how these exposures contribute to cancer incidence across countries and regions have been limited. This study addresses such gap, providing a robust evidence base to guide prioritization of cancer prevention strategies and inform population-level policy action.

The solution

We conducted a comparative risk assessment to quantify how much of the global cancer incidence in 2022 could be attributed to modifiable risk factors. We used incidence estimates for 36 types of cancers across 185 countries from the GLOBOCAN open-source database and combined these data with harmonized prevalence estimates for risk-factor exposures drawn from major international surveillance systems and scientific literature. Thirty risk factors were included, selected on the basis of established causal evidence and data availability. To account for the time between risk-factor exposure and cancer development, we used exposure prevalence data from ten years earlier (around 2012). Population-attributable fractions (the proportions of disease cases that can be attributed to a specific exposure) were calculated using established methods based on Levin's or Miettinen's formulas[5] or direct attribution for necessary or sufficient causes. Importantly, we accounted for overlapping exposures to estimate their combined attributable burden by cancer types, sex, country and region, ensuring comparability and policy relevance.

We found that around four in ten cancers globally – corresponding to 7.1 million new cases in 2022 – were attributable to the combined effect of modifiable risk factors, with a substantially higher attributable burden in men than in women. This disparity largely reflects higher exposure among men to key risk factors, particularly tobacco smoking and alcohol consumption, which together account for a much larger share of cancer cases in men across most regions. By contrast, the distribution of attributable risk in women was more heterogeneous, with infections, high body mass index (BMI) and smoking contributing differently across regions. The proportion of attributable cancers varied widely across regions, reflecting differences in dominant exposures and stages of epidemiological transition – shifts in population risk profiles, from infection-related cancers (driven mainly by *Helicobacter pylori*, human papillomavirus, and hepatitis B and C viruses) in lower-income settings to cancers linked to behavioral risk factors such as tobacco smoking, alcohol consumption and high BMI in higher-income settings (Fig. 1). High BMI, air pollution and ultraviolet radiation played an important part in specific regions. Lung, stomach and cervical cancers together accounted for nearly half of the attributable burden.

The implications

The strong regional and sex-specific patterns observed highlight the need for prevention strategies that are tailored to local exposure profiles, health system capacity and social context. By quantifying the attributable cancer burden across countries, this work provides actionable evidence to support prioritization of cancer prevention within national cancer control plans and reinforces prevention as a cornerstone of global cancer control.

This study relies on the quality and completeness of available data on cancer incidence, exposure prevalence and risk estimates, which remain uneven across regions, particularly in low- and middle-income countries. Estimates assume broadly consistent risk estimates across populations and apply a harmonized latency period, which might not fully capture variation in exposure–disease relationships. In addition, uncertainty estimates could not be applied consistently across all risk factors and settings owing to a lack of comparable variance information; interactions between co-occurring exposures were not explicitly modelled.

Future work should incorporate improved surveillance data, region-specific risk estimates and dynamic exposure histories to better reflect real-world complexity. Extending this framework to model trends over time and evaluate the potential effect of prevention scenarios would further strengthen its value for policy and planning.

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EXPERT OPINION

"The findings of this study can serve as a critical guide for governments, policymakers and healthcare providers to prioritize economic investments and design targeted cancer prevention and control programs. Notably, the identification of infections as a major modifiable risk factor for cancer — an area often underemphasized in previous global analyses — is a key contribution of this paper. Overall, this study makes a valuable addition to the literature and provides an actionable framework for guiding national and global cancer prevention strategies."
Shilpa S. Murthy, Yale University, New Haven, CT, USA.

FIGURE

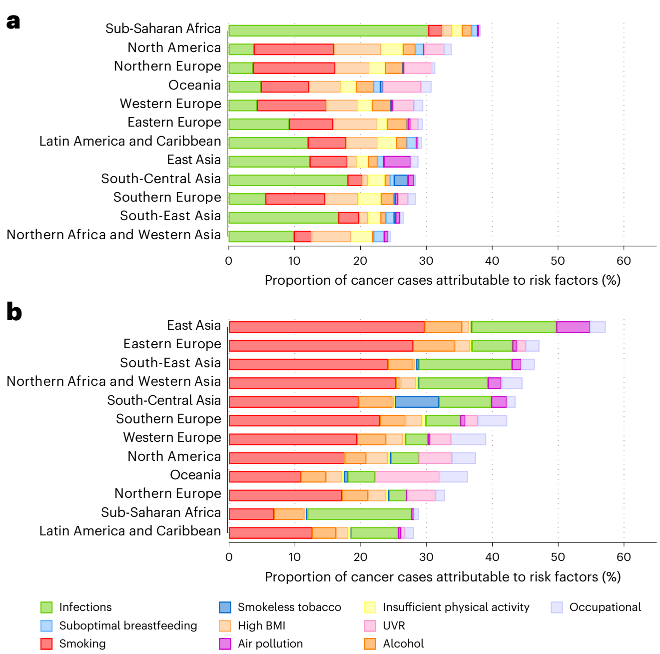

Fig. 1 | Burden of cancer due to modifiable risk factors by sex and region. a,b, Proportions of cancer cases attributable to selected modifiable risk factors across world regions in 2022 in women (**a**) and men (**b**). Risk-factor exposures result in marked regional and sex-specific differences in cancer burden. Smokeless tobacco refers to non-combusted tobacco products such as chewing tobacco, snuff, snus and region-specific preparations used orally. UVR, ultraviolet radiation. © 2026, Fink, H. et al.

BEHIND THE PAPER

This project began with a deceptively simple question: how much of today's global cancer burden could potentially be prevented? Turning that question into a global analysis across 185 countries in just one year was both exhilarating and daunting. I am deeply grateful to my supervisor, I.S., for trusting me with such an ambitious project at an early stage of my PhD, despite the late nights, moments of doubt and more than a few tears along the way. The collaboration behind this work was extraordinary: our co-authors were consistently engaged, generous with their expertise and essential at every step. This study would not exist without our data manager, Jérôme Vignat, whose meticulous work cleaning and harmonizing complex datasets made the analysis possible. Amid the intensity, it was genuinely fun to explore patterns, refine messages and uncover striking regional differences. I feel incredibly honored to contribute, in some small way, to cancer prevention, because no one should suffer from this disease.**H.F.**

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FROM THE EDITOR

"By assessing cancer burden attributable to different risk factors specific to each region, countries can plan and prioritize prevention programs suited to their own needs."
Editorial Team, *Nature Medicine*.